

2 (long)

(a) Describe the advantages, for a circuit designer, of *negative feedback* on an amplifier. [6]

(b) Draw a labelled equivalent circuit of an Op Amp with an input resistance of R_{in} , an output resistance of R_{out} and a gain of A . [4]

(c) Figure 2 shows an Op Amp circuit driving a cable of resistance R_3 , and a load of resistance R_4 in parallel with capacitance C . Taking $A = 10^4$, $R_1 = 4.5 \text{ k}\Omega$, $R_2 = 500 \Omega$, $R_3 = 75 \Omega$, $R_4 = 150 \text{ k}\Omega$ and $C = 100 \text{ pF}$, calculate the approximate mid-band gain and 3dB frequency of the circuit. [20]

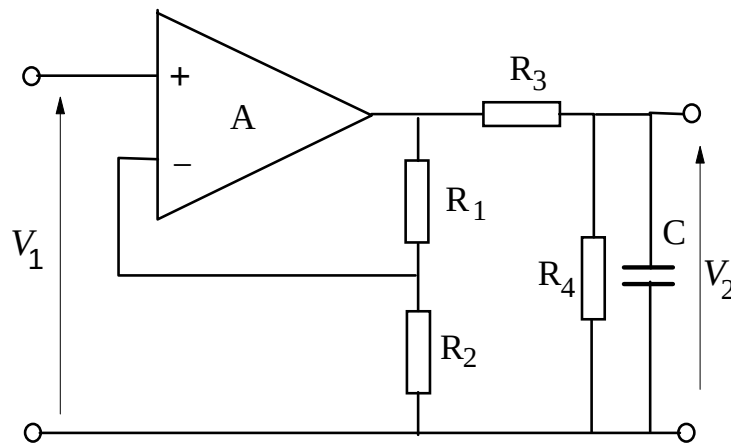


Fig. 2